

Negative prime-position powers do not define a p-adic number

CONJECTURE 00000000714 | FALSE AS WRITTEN (DIVERGENT SERIES)

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Norm obstruction

In $\mathbb{Q}_p$, $|p^{(-p_n)}|_p = p^{(p_n)}$, which is at least 1 for every n . The terms never approach zero.

Divergence

Terms of a convergent series must approach zero, being differences of successive partial sums. Hence this series has no p-adic sum and cannot define the claimed transcendental number.

Scope

The minus sign is essential. Replacing negative exponents by positive exponents changes the object, and that corrected series is not addressed.

Formalization

Lean uses the actual norm on $\text{Padic } p$, proves the obstruction for every natural-exponent sequence, and proves non-summability for Nat.nth Nat.Prime . This is a convergence obstruction, not a numerical test or a claim about the corrected positive-exponent series.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000714.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.